Assignment 1

richardsuwandi@link.cuhk.edu.cn

Please note that

- **Released date: January 16, Tuesday.**
- **Due date: January 27, Saturday, by 11:55pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "123010XXX ASS1.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Label the following statements as true or false:

(a) If a system of linear equations has two different solutions, it must have infinitely many solutions.

(b) If B is a matrix that can be obtained by performing an elementary row operation on a matrix A, then A can be obtained by performing an elementary row operation on B.

(c) Every matrix can be transformed into a unique row echelon form using elementary row operations.

Solution

(a) true (b) true (c) false (REF is not unique; RREF is unique)

Question 2. (a) Write down the coefficient matrix and the augmented matrix for the linear system

$$\begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases}.$$

(b) Solve the system using Gauss-Jordan elimination.

Solution

(a) The coefficient matrix is

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 2 & -4 & 1 & 0 \\ 1 & -2 & 2 & -3 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right]$$

(b) By Gauss–Jordan elimination,

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xrightarrow[R_3 \to -R_1 + R_3]{R_2 \to -2R_1 + R_2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_2 + R_3} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to \frac{1}{3}R_2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_2 + R_1} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This is equivalent to the linear system

$$\begin{cases} x_1 - 2x_2 + x_4 = 2 \\ x_3 - 2x_4 = 1. \end{cases}$$

The solutions are given by

$$\begin{cases} x_1 = 2 + 2x_2 - x_4 \\ x_3 = 1 + 2x_4, \end{cases}$$

where x_2 and x_4 are free variables which can be arbitrary real numbers.

Question 3. Determine whether the following linear systems are equivalent or not:

$$1)\begin{cases} x-y+2z=1 \\ 2x+y+z=8 \\ x+y=5 \end{cases} \quad 2)\begin{cases} x+y=5 \\ x+z=3 \\ y-z=2 \end{cases}.$$

Solution

We apply Gauss–Jordan elimination to system 1 to get the RREF:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & 1 & 1 & 8 \\ 1 & 1 & 0 & 5 \end{array}\right] \to \cdots \to \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Applying Gauss–Jordan elimination to system 2 yields

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \end{array}\right] \to \cdots \to \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

They both have the same reduced row echelon form (RREF), so the two systems are equivalent.

Question 4. Which of the following matrices are in row echelon form? Which are in reduced row echelon form?

(a) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ (f) $\begin{bmatrix} 1 & 4 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ (g) $\begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 2 & 4 \\ 0 & 0 & 1 & 3 & 6 \end{bmatrix}$ (h) $\begin{bmatrix} 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Solution

REF: (a), (c), (d), (e), (g), (h)
RREF: (c), (d), (g)

Question 5. Find a condition on the numbers a, b, and c such that the following system of equations is consistent. When that condition is satisfied, find all solutions (in

terms of a, b, and c).

$$\begin{cases} x_1 + 3x_2 + x_3 = a \\ -x_1 - 2x_2 + x_3 = b \\ 3x_1 + 7x_2 - x_3 = c \end{cases}$$

Solution

First compute REF:

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ -1 & -2 & 1 & b \\ 3 & 7 & -1 & c \end{array}\right] \xrightarrow[R_3 \to -3R_1 + R_3]{R_2 \to R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & -2 & -4 & -3a+c \end{array}\right]$$

$$\xrightarrow{R_3 \to 2R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & -2a+b+c \end{array}\right].$$

Hence, the system is consistent if and only if the condition $-2a + b + c = 0$ holds. When this condition holds, we further compute the RREF:

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -3R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & 0 \end{array}\right].$$

From the RREF, we see that the solutions are

$$\begin{cases} x_1 = -2a - 3b + 5x_3 \\ x_2 = a + b - 2x_3 \end{cases}$$

where x_3 is a free variable which is an arbitrary real number.

Question 6. Consider the linear system

$$\begin{cases} x + 3y - 2z = 1 \\ x + 4y + z = 5 \\ 2y + pz = q \end{cases}.$$

(a) Find the values of p and q so that the system has a unique solution.

(b) Find the values of p and q so that the system has no solution.

(c) Find the values of p and q so that the system has infinitely many solutions, and then find the solution set.

Solution

First compute REF:

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 1 & 4 & 1 & 5 \\ 0 & 2 & p & q \end{array}\right] \xrightarrow{R_2 \to -R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & p & q \end{array}\right]$$

$$\xrightarrow{R_3 \to -2R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & p-6 & q-8 \end{array}\right].$$

(a) The system has a unique solution if and only if $p \neq 6$, and q can be any real number.

(b) The system has no solution if and only if $p = 6$ and $q \neq 8$.

(c) The system has infinitely many solutions if and only if $p = 6$ and $q = 8$. When this is the case, the RREF is

$$\left[\begin{array}{ccc|c} 1 & 0 & -11 & -11 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The solution is

$$\begin{cases} x = -11 + 11z \\ y = 4 - 3z \end{cases}$$

where z is an arbitrary real number. In other words, the solution set is

$$\left\{ \begin{bmatrix} -11 + 11t \\ 4 - 3t \\ t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Question 7. Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array}\right].$$

For what values of β will the system have non-trivial solutions?

Solution

We compute REF as follows:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array}\right] \xrightarrow[R_3 \to R_1 + R_3]{R_2 \to -R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 3 & \beta+1 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \to -3R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & \beta-2 & 0 \end{array}\right].$$

Hence, the system has non-trivial solutions if and only if $\beta = 2$.

Question 8. Consider the linear system $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 4 & 4 \\ 2 & -2 & a \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix}.$$

(a) For what values of a will the system have a unique solution?

(b) Suppose $a = 1$. Write $\mathbf{b}$ as a linear combination of the column vectors of A.

Solution

(a) We compute REF as follows:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ -1 & 4 & 4 & -2 \\ 2 & -2 & a & 7 \end{array}\right] \xrightarrow[R_3 \to -2R_1 + R_3]{R_2 \to R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 5 & -1 \\ 0 & -6 & a-2 & 5 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 5 & -1 \\ 0 & 0 & a+3 & 4 \end{array}\right].$$

Hence, the system has a unique solution if and only if $a \neq -3$.

(b) When $a = 1$, the REF is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 5 & -1 \\ 0 & 0 & 4 & 4 \end{array}\right].$$

By back substitution, we obtain the solution $x_1 = 2, x_2 = -1, x_3 = 1$.

Hence, $\mathbf{b}$ can be written as the following linear combination:

$$\mathbf{b} = 2\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}.$$

Question 9. For each of the following pairs of matrices, determine whether it is possible to multiply the first matrix times the second. If it is possible, perform the multiplication.

(a) $\begin{bmatrix} 3 & 5 & 1 \\ -2 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & -2 \\ 6 & -4 \\ 8 & -6 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 5 \end{bmatrix}$

(d) $\begin{bmatrix} 4 & 6 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{bmatrix}$ (e) $\begin{bmatrix} 4 & 6 & 1 \\ 2 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{bmatrix}$ (f) $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 & 2 & 4 & 5 \end{bmatrix}$

Solution

(a) $\begin{bmatrix} 15 & 19 \\ 4 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 19 & 21 \\ 17 & 21 \\ 8 & 10 \end{bmatrix}$ (d) $\begin{bmatrix} 36 & 10 & 56 \\ 10 & 3 & 16 \end{bmatrix}$ (f) $\begin{bmatrix} 6 & 4 & 8 & 10 \\ -3 & -2 & -4 & -5 \\ 9 & 6 & 12 & 15 \end{bmatrix}$

(b) and (e) are not possible

Question 10. Give an example of 2×2 matrices A and B such that $AB = O$ but $BA \neq O$.

Solution

For example,

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow AB = O,\ BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Question 11. One may raise a square matrix to any non-negative integer power multi-

plying it by itself repeatedly in the same way as for ordinary numbers, that is,

$$\begin{aligned} A^0 &= I, \\ A^1 &= A, \\ A^2 &= AA, \\ A^k &= \underbrace{AA\cdots A}_{k \text{ times}}. \end{aligned}$$

Suppose

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

calculate the matrix A^k.

Solution

Note that $A^2 = I_3$, and thus $A^3 = A^2A = A$. Therefore, by induction, we conclude that for any integer $k \geq 1$,

$$A^k = \begin{cases} A & \text{if } k \text{ is odd,} \\ I_3 & \text{if } k \text{ is even.} \end{cases}$$

Question 12. Let

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Compute A^2 and A^3. What will A^{2n} and A^{2n+1} turn out to be?

Solution

Direct computation yields $A^2 = I_4$, and thus $A^3 = A^2A = A$. Therefore, $A^{2n} = I_4$ and $A^{2n+1} = A$.

Question 13. Suppose the size of the following matrices are 2×2, give an example to show that the following results are false:

(a) $A^2 = O$, then $A = O$.

(b) $A^2 = A$, then $A = O$ or $A = I$.

Solution

(a) For example,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

is not the zero matrix, but we still have $A^2 = O$.

(b) For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

is neither O nor I, but we still have $A^2 = A$.

Question 14. Is it possible for a nonzero symmetric 2×2 matrix to have the property that $A^2 = O$? Prove your answer.

Solution

Updated solution: This question is not clear enough since the answer depends on whether the matrix is real or complex. For nonzero symmetric 2×2 real matrix, the answer is no. Proof: Any 2×2 symmetric matrix is of the form

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}.$$

It follows that

$$A^2 = \begin{bmatrix} a^2 + b^2 & ab + bc \\ ab + bc & b^2 + c^2 \end{bmatrix}.$$

If A is a real matrix and $A^2 = O$, then its diagonal entries must be 0, i.e., $a^2 + b^2 = 0$ and $b^2 + c^2 = 0$. But a sum of nonnegative numbers can be 0 only if all terms are 0, so we have $a = b = c = 0$, i.e., A must be the zero matrix. Hence, it is not possible for a nonzero 2×2 symmetric real matrix to have the property that $A^2 = O$.

For nonzero symmetric 2×2 complex matrix, the answer is yes. We can take

$$A = \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}$$

which is symmetric but $A^2 = O$.

Question 15. Let A be a 3×3 matrix whose entries on the diagonal and below the diagonal are all 0. Show that $A^3 = O$.

Solution

Such a matrix is of the form

$$A = \begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix}.$$

It is straightforward to show that

$$A^2 = \begin{bmatrix} 0 & 0 & ac \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and $A^3 = O$.

Question 16. Prove that if A is a real square matrix, and $A^\top A = O$, then $A = O$.

Solution

Let $A = (a_{ij})_{n\times n}$. For all $i \in \{1, \ldots, n\}$, the (i, i)-th entry of $A^\top A$ is

$$(A^\top A)_{ii} = \sum_{k=1}^{n} (A^\top)_{ik} A_{ki} = \sum_{k=1}^{n} a_{ki}^2.$$

Therefore, $A^\top A = O$ implies that $\sum_{k=1}^{n} a_{ki}^2 = 0$ for all $i \in \{1, \ldots, n\}$. But a sum of nonnegative numbers can be 0 only if all terms are 0, so $\sum_{k=1}^{n} a_{ki}^2 = 0$ implies that $a_{ki} = 0$ for all $k \in \{1, \ldots, n\}$. Therefore, $a_{ki} = 0$ for all i and all k. This means that $A = O$.

Question 17. A square matrix A is said to be idempotent if $A^2 = A$.

Let A and B be two symmetric idempotent $n \times n$ matrices. Show that $AB = B$ if and only if $A - B$ is idempotent.

Solution

($\Rightarrow$). Suppose $AB = B$. First, note that

$$(A - B)^2 = (A - B)(A - B) = A^2 - AB - BA + B^2.$$

Since A and B are both idempotent and $AB = B$, the above simplifies to

$$(A - B)^2 = A - BA. \tag{1}$$

Moreover, thanks to the properties that A and B are both symmetric and

$AB = B$, we deduce that

$$BA = (A^\top B^\top)^\top = (AB)^\top = B^\top = B.$$

Putting this back into (1) shows that $(A - B)^2 = A - B$, that is, $A - B$ is idempotent.
($\Leftarrow$). Conversely, if $A - B$ is idempotent, then

$$A - B = (A - B)^2 = A - AB - BA + B,$$

which simplifies to

$$2B = AB + BA. \tag{2}$$

Multiplying both sides of (2) by B from the left and using the idempotent property of B, we obtain

$$2B = BAB + BA. \tag{3}$$

Similarly, multiplying both sides of (2) by B from the right yields

$$2B = AB + BAB. \tag{4}$$

Now (3) and (4) together imply that $AB = BA$. Putting this back into (2), we get that $2B = 2AB$ and hence $AB = B$.